

Deep Inelastic Scattering Revisited

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Abstract

Discuss Yadism [\[1\]](#).

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Chapter 1

Introduction

1.1 Introduction

- Ich + paper + yad
- Ziele und $\neg$ Ziele
- D I S
- factorization theorem
- LO (+ NLOq + NLOg)
- IC

1.2 SF and xs

- x, Q^2
- $L_{\mu\nu} W^{\mu\nu}$

1.3 RSL

Chapter 2

DIS at next level

2.1 More particles

- γ/Z
- leptons
- hadrons: $p+d+A$; nucl. PDF
- experimental overview

2.2 DGLAP

- resummation
- Mellin

2.3 Scale variations

- scheme A/B/C
- MHOU
- covariance matrix

2.4 Heavy Quarks

- F_2^c
- FONLL

2.5 γ_5

- helicity
- HVBM + Larin

2.6 Positivity

- DIS vs. MSbar

2.7 Higher orders

- IHOU
- N3LO
- channels
- loops and legs

2.8 QED

- Qiu et al.

2.9 Valo

- $C_\gamma \rightarrow k$

Bibliography

- [1] A. Candido, F. Hekhorn, G. Magni, T. R. Rabemananjara, and R. Stegeman, Eur. Phys. J. C **84**, 698 (2024), 2401.15187.